Gaussian states, covariance matrices and the formal Fock space

Gaussian formalism expresses properties of bosonic quantum states in terms of phase-space data. For Gaussian states, the first moments and covariance matrix determine the state: the former give its mean position and momentum, and the latter describe its fluctuations. These notes develop the matrix algebra of covariances and connect it with explicit normalized Gaussian wave functions, laying foundations for the study of Gaussian matrix product states. For general background, see Weedbrook et al. [1, Sec. II].

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About these notes

These notes are the mathematical account of the Lean library BosonicLean. We assume finite-dimensional linear algebra, complex inner products, Lebesgue integration and elementary differential calculus; Section 6 needs only algebra and formal power series. The reading guide below relates our conventions to those of the literature, the roadmap after it shows which sections use which, and an index of terms precedes the references.

A statement marked Lean ✓ has a checked proof in BosonicLean whose correspondence with the statement here has been reviewed; the declaration index at the end names the checked declarations. Definitions, Remarks and paragraphs of motivation carry no marks.

Reading guide

Where our objects or conventions differ from the usual physics presentation, the table names the object used here, the one used in the literature, and where the two are related.

Here	**In the literature**	**Relation**
Covariance γ, twice the symmetrized covariance; vacuum I_2	Varies: vacuum I_2 as here (Weedbrook et al. [1], quadratures $a + a^\dagger$); $\sigma = \gamma/2$ with vacuum $\frac{1}{2}I_2$ when σ is the symmetrized covariance itself; or $\hbar$-dependent units	Definition 2.3; units in Section 2
Coordinates $(q_1, \ldots, q_n, p_1, \ldots, p_n)$, form Ω_n	Mode by mode, $(q_1, p_1, \ldots, q_n, p_n)$ (e.g. [1])	A permutation of coordinates carries one symplectic form to the other; covariances transform by the same permutation, by congruence, and admissibility is unchanged (proof of Theorem 1.5)
Admissibility $\gamma - i\Omega_n \succeq 0$	Uncertainty relation for the covariance, e.g. $V + i\Omega \succeq 0$	Definition 1.1; equivalent to $\gamma + i\Omega_n \succeq 0$ by complex conjugation, and for $\sigma = \gamma/2$ to $\sigma - \frac{i}{2}\Omega_n \succeq 0$
Schwartz functions inside $L^2(\mathbb{R})$	Wave functions	Section 2
Contraction: a Schur complement of covariance matrices	Projection of virtual modes onto EPR pairs	Section 3: the ideal EPR vector is not normalizable; the Schur complement is proved, the EPR projection is its motivation
Finite periodic chains, $N \geq 3$ (translation-invariant in Section 4)	Often infinite chains	Sections 3–5
Formal Fock space: all coefficient families	The Hilbert space $\Gamma_s(\mathfrak{h})$	$\Gamma_s(\mathfrak{h})$ is the square-summable part (Section 6)
e^T as a finite sum at each coefficient, for T raising the particle number	Operator exponential	Equal coefficientwise wherever the series converges (after Lemma 6.5)
Bargmann map into formal power series	Bargmann's space of entire functions	On $\Gamma_s(\mathfrak{h})$, the Taylor coefficients (after Definition 6.7)

Roadmap

Section 1 characterizes covariance matrices by the matrix inequality (1.1) and proves that symplectic congruence, the selection of a subsystem and independent sums preserve it. Section 2 constructs normalized Gaussian wave functions and computes their covariances, which realize the equality case of the one-mode criterion of Proposition 1.6. Section 3 defines the contraction of covariance matrices, a Schur complement motivated by EPR measurements on virtual modes, proves that it preserves admissibility, and assembles from it the covariance of a periodic Gaussian matrix product state. Sections 4 and 5 treat two cases of that construction: chains with the same local covariance at every site, whose covariance splits into one 2×2 block per frequency, and products.

Section 6 uses none of Sections 1–5. From algebra and formal power series alone it builds the formal Fock space of all coefficient families over the number basis, its ladder operators, the exponentials of operators that raise the particle number, and the Bargmann picture; it is the foundation of the Gaussian-augmented bosonic matrix product states of [4]. Figure 1 shows which sections use which.

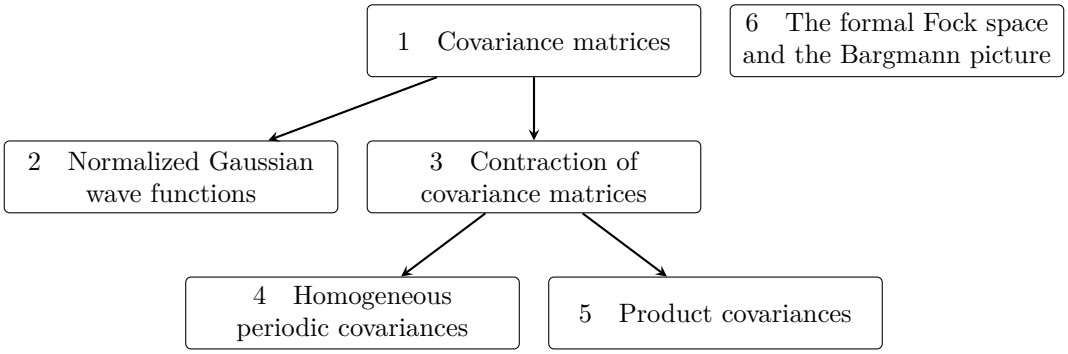


Figure 1: Dependence of the sections. An arrow means that the later section uses the earlier one; arrows implied by a chain of arrows are omitted. Section 2 enters Sections 3 and 5 only in remarks, and Section 6 uses no other section.

1 Covariance matrices

For $n \geq 0$, we use coordinates $(q_1, \ldots, q_n, p_1, \ldots, p_n)$ on $\mathbb{R}^{2n}$ and define

$$\Omega_n = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

For a complex matrix H, the notation $H \succeq 0$ means that $H = H^*$ and $z^* H z \geq 0$ for every complex column vector z. Here H^* is the conjugate transpose. We regard real matrices as complex matrices in this inequality.

Definition 1.1 (Admissible covariance)**.** A real $2n \times 2n$ matrix γ is called *admissible* if

$$\gamma^T = \gamma, \qquad \gamma - i\Omega_n \succeq 0. \tag{1.1}$$

These coordinates are dimensionless. For a real $2n \times 2n$ matrix γ we call $\gamma - i\Omega_n$ its *uncertainty matrix*. For one mode the condition concerns the Hermitian matrix

$$\begin{pmatrix} a & c - i \\ c + i & b \end{pmatrix}, \qquad \gamma = \begin{pmatrix} a & c \\ c & b \end{pmatrix}.$$

The imaginary part records the canonical pairing between position and momentum. The empty matrix is admissible when $n = 0$.

1.1 Changes of coordinates

A real matrix S is *symplectic* in these coordinates when $S\Omega_n S^T = \Omega_n$. Transforming a covariance in both indices gives $\gamma \mapsto S\gamma S^T$. This is a *congruence transformation*. Similarity has the form $A \mapsto SAS^{-1}$; the two formulas agree for orthogonal S, but symplecticity alone does not give $S^T = S^{-1}$.

Theorem 1.2. Lean ✓ *Let $n \geq 0$ and let S, γ be real $2n \times 2n$ matrices. If γ is admissible and $S\Omega_n S^T = \Omega_n$, then $S\gamma S^T$ is admissible.*

Proof. Symmetry follows from $(S\gamma S^T)^T = S\gamma^T S^T = S\gamma S^T$. Put $H = \gamma - i\Omega_n$. Since S is real,

$$S\gamma S^T - i\Omega_n = S(\gamma - i\Omega_n)S^T = SHS^*.$$

This matrix is Hermitian, and for every $z \in \mathbb{C}^{2n}$,

$$z^* SHS^* z = (S^* z)^* H (S^* z) \geq 0.$$

The same identities hold for empty matrices. □

1.2 Subsystems and independent sums

To select a subsystem, a position coordinate and its matching momentum coordinate must be selected together. A *one-to-one map*, also called an *injective map* or an *injection*, is a map e such that $e(i) = e(j)$ implies $i = j$. Thus a one-to-one map $e : \{1, \ldots, m\} \to \{1, \ldots, n\}$ selects distinct modes without repetitions, in the order

$$(q_{e(1)}, \ldots, q_{e(m)}, p_{e(1)}, \ldots, p_{e(m)}).$$

Write $E(q_j) = q_{e(j)}$ and $E(p_j) = p_{e(j)}$. The selected covariance has entries $\gamma'_{ab} = \gamma_{E(a),E(b)}$.

Theorem 1.3. Lean ✓ *For all $m, n \geq 0$, every such selection from an admissible n-mode covariance is an admissible m-mode covariance. The empty selection is included.*

Proof. Symmetry is inherited by restriction. The qq and pp blocks of Ω_n vanish, and its qp and pq entries are δ_{ij} and $-\delta_{ij}$. Injectivity gives $\delta_{e(i),e(j)} = \delta_{ij}$, so restriction along E sends Ω_n to Ω_m.

Thus $H' = \gamma' - i\Omega_m$ is the selected principal submatrix of $H = \gamma - i\Omega_n$. It is Hermitian. Extending a vector $z \in \mathbb{C}^{2m}$ by zero on the unselected coordinates gives $\widetilde{z} \in \mathbb{C}^{2n}$ and

$$z^* H' z = \widetilde{z}^* H \widetilde{z} \geq 0.$$

□

Conversely, we can combine independent covariance blocks without cross terms.

Definition 1.4 (Independent sum)**.** For arbitrary real matrices A and B of sizes $2n \times 2n$ and $2m \times 2m$, write their position/momentum blocks as $A_{qq}, A_{qp}, A_{pq}, A_{pp}$ and similarly for B. Define

$$A \boxplus B = \begin{pmatrix} A_{qq} & 0 & A_{qp} & 0 \\ 0 & B_{qq} & 0 & B_{qp} \\ A_{pq} & 0 & A_{pp} & 0 \\ 0 & B_{pq} & 0 & B_{pp} \end{pmatrix} \tag{1.2}$$

in the order (q_A, q_B, p_A, p_B). This is $\mathrm{diag}(A, B)$ reordered from (q_A, p_A, q_B, p_B); the definition makes no symmetry assumption.

Theorem 1.5. Lean ✓ *For all $n, m \geq 0$ and all real matrices A, B of the stated sizes, $A \boxplus B$ is admissible if and only if both A and B are admissible.*

Proof. The same coordinate permutation sends $\operatorname{diag}(\Omega_n, \Omega_m)$ to Ω_{n+m}. Hence the uncertainty matrix of $A \boxplus B$ is the corresponding permutation of $\operatorname{diag}(A - i\Omega_n, B - i\Omega_m)$. When both blocks are admissible, symmetry is preserved and the quadratic form is the sum of two nonnegative quadratic forms. Hermiticity is inherited from the blocks. Conversely, restriction to either collection of matching position/momentum coordinates recovers that block and its uncertainty matrix. Theorem 1.3 applies. Empty blocks cause no exception. □

1.3 One-mode diagonal matrices

The diagonal case gives explicit examples and fixes the scale of the uncertainty bound. Put $D(a, b) = \operatorname{diag}(a, b)$ in the order (q, p).

Proposition 1.6. Lean ✓ *For all $a, b \in \mathbb{R}$, the matrix $D(a, b)$ is admissible if and only if $a > 0$ and $ab \geq 1$.*

Proof. The uncertainty matrix is $H = \left(\begin{smallmatrix} a & -i \\ i & b \end{smallmatrix}\right)$. If $H \succeq 0$, the coordinate vectors give $a, b \geq 0$. If $a = 0$, the vector $(b + 1, -i)^T$ gives the negative value $-b - 2$; hence $a > 0$. The vector $(i, a)^T$ then gives $0 \leq a(ab - 1)$, and therefore $ab \geq 1$.

For the converse, suppose $a > 0$ and $ab \geq 1$. Write $z = x + iy$, $w = u + iv$. The quadratic form is real and satisfies

$$\begin{aligned} Q &= (\overline{z}, \overline{w}) H (z, w)^T = a(x^2 + y^2) + b(u^2 + v^2) + 2(xv - yu), \\ aQ &= (ax + v)^2 + (ay - u)^2 + (ab - 1)(u^2 + v^2) \geq 0. \end{aligned}$$

Thus $Q \geq 0$; Hermiticity of H is immediate. □

In particular, $I_2 = D(1, 1)$ is admissible. More generally, $D(t, t^{-1})$ is admissible for every $t > 0$. Its two variances in our covariance convention are $t/2$ and $1/(2t)$; their product is $1/4$. By contrast, cI_2 with $0 \leq c < 1$ is a positive semidefinite real matrix but is not admissible, since $c^2 < 1$. We now construct wave functions whose covariances give the one-mode equality case.

2 Normalized Gaussian wave functions

Definition 2.1 (Schwartz space)**.** Let $\mathcal{D} = \mathcal{S}(\mathbb{R}; \mathbb{C})$ be the Schwartz space: a smooth function f belongs to $\mathcal{D}$ when

$$\sup_{x \in \mathbb{R}} |x|^r |f^{(k)}(x)| < \infty \qquad (r, k \in \mathbb{N}_0).$$

We use Lebesgue measure and put $\mathcal{H} = L^2(\mathbb{R}; \mathbb{C})$. Elements of $\mathcal{H}$ are equivalence classes of functions: two functions represent the same element if they agree almost everywhere. Schwartz decay makes every $f \in \mathcal{D}$ square-integrable. The usual inclusion

$$J : \mathcal{D} \longrightarrow \mathcal{H}, \qquad Jf = [f],$$

sends a Schwartz function to its class. It is injective: continuous functions that agree almost everywhere agree everywhere. We henceforth identify f with Jf when taking an L^2 norm or inner product, writing

$$\|f\|_2 = \|Jf\|_{\mathcal{H}}, \qquad \langle f, g \rangle = \int_{\mathbb{R}} \overline{f(x)} g(x)\, dx.$$

The inner product is conjugate-linear in its first argument.

Definition 2.2 (Normalized representative)**.** A normalized state representative is a Schwartz function $f \in \mathcal{D}$ with $\|f\|_2 = 1$.

For $\hbar > 0$, position and momentum act on the common invariant domain $\mathcal{D}$ by

$$(Xf)(x) = xf(x), \qquad (P_\hbar f)(x) = -i\hbar f'(x).$$

Multiplication by x and differentiation preserve Schwartz space. Their images therefore have well-defined L^2 norms and inner products.

Definition 2.3 (Means and covariance)**.** For an operator $A : \mathcal{D} \to \mathcal{D}$ and a normalized f, define

$$\mu_f(A) = \mathrm{Re}\langle f, Af\rangle, \qquad u_f(A) = Af - \mu_f(A)f.$$

For the ordered pair $(R_1, R_2) = (X, P_\hbar)$, the covariance is

$$(\gamma_f)_{jk} = 2\,\mathrm{Re}\langle u_f(R_j), u_f(R_k)\rangle. \tag{2.1}$$

In particular, $(\gamma_f)_{jj} = 2\|u_f(R_j)\|_2^2$ is twice the variance.

2.1 Existence and normalization

For a width $s > 0$, we consider the positive real function

$$\psi_s(x) = \frac{1}{\sqrt{\sqrt{\pi}\, s}} \exp\left(-\frac{x^2}{2s^2}\right). \tag{2.2}$$

Theorem 2.4. Lean ✓ *For every $s > 0$, the function ψ_s belongs to $\mathcal{D}$ and satisfies $\|\psi_s\|_2 = 1$. It determines a unique normalized state representative with the pointwise values in* (2.2)*.*

Proof. Put $\phi(x) = e^{-x^2/2}$. We define the *Hermite polynomials* by

$$H_0(x) = 1, \qquad H_{k+1}(x) = xH_k(x) - H_k'(x) \quad (k \geq 0).$$

The *Hermite derivative identity* is

$$\phi^{(k)}(x) = (-1)^k H_k(x)\phi(x).$$

It follows by induction from $\phi' = -x\phi$ and the product rule. For every $r \geq 0$, $|x|^r\phi(x) \to 0$ as $|x| \to \infty$. Continuity bounds this function globally. Expanding H_k into monomials then bounds $|x|^r|\phi^{(k)}(x)|$ for each r, k. Dilation by $x \mapsto x/s$, inclusion of real values into $\mathbb{C}$, and multiplication by a constant preserve Schwartz space, so $\psi_s \in \mathcal{D}$.

The Gaussian integral gives

$$\|\psi_s\|_2^2 = \frac{1}{\sqrt{\pi}\, s} \int_{\mathbb{R}} e^{-x^2/s^2}\, dx = \frac{\sqrt{\pi}\, s}{\sqrt{\pi}\, s} = 1.$$

A function with the prescribed pointwise values is equal to ψ_s. □

The pointwise formula (2.2) fixes the overall phase of ψ_s.

2.2 Action and moments

Theorem 2.5. Lean ✓ *For every $s, \hbar > 0$,*

$$P_\hbar\psi_s = \frac{i\hbar}{s^2} X\psi_s, \qquad \mu_{\psi_s}(X) = \mu_{\psi_s}(P_\hbar) = 0, \tag{2.3}$$

$$\|X\psi_s\|_2^2 = \frac{s^2}{2}, \qquad \|P_\hbar\psi_s\|_2^2 = \frac{\hbar^2}{2s^2}, \tag{2.4}$$

$$\langle X\psi_s, P_\hbar\psi_s\rangle = \frac{i\hbar}{2}. \tag{2.5}$$

The first identity holds as an equality of Schwartz functions.

Proof. Differentiating (2.2) yields $\psi_s' = -(x/s^2)\psi_s$ and hence the momentum action. The integrand in $\langle \psi_s, X\psi_s \rangle$ is odd and integrable, so this complex inner product vanishes.

Set $A = \|X\psi_s\|_2^2$. With $\langle \cdot, \cdot \rangle_{\mathbb{R}}$ denoting the real part of the complex inner product, integration by parts gives

$$1 - \frac{A}{s^2} = \langle \psi_s, (X\psi_s)' \rangle_{\mathbb{R}} = -\langle \psi_s', X\psi_s \rangle_{\mathbb{R}} = \frac{A}{s^2}. \tag{2.6}$$

The first equality uses $(X\psi_s)' = \psi_s + x\psi_s'$ and normalization. Schwartz decay removes the boundary term, and the derivative formula gives the last equality. Thus $A = s^2/2$.

Complex linearity in the second argument now gives

$$\langle \psi_s, P_\hbar \psi_s \rangle = 0, \qquad \langle X\psi_s, P_\hbar \psi_s \rangle = \frac{i\hbar}{s^2} A = \frac{i\hbar}{2}.$$

Finally, $\|P_\hbar \psi_s\|_2^2 = (\hbar^2/s^4)A = \hbar^2/(2s^2)$. □

2.3 Covariance and units

Theorem 2.6. Lean ✓ *For every $s, \hbar > 0$, the covariance of ψ_s is*

$$\gamma_{\psi_s} = \begin{pmatrix} s^2 & 0 \\ 0 & \hbar^2/s^2 \end{pmatrix}.$$

If $T = \operatorname{diag}(s^{-1}, s/\hbar)$ and $\widetilde{\gamma} = T\gamma_{\psi_s}T^T$, then $\widetilde{\gamma} = I_2$ and $\widetilde{\gamma}$ is admissible in the sense of (1.1).

Proof. The means vanish, so the centered vectors in (2.1) are $X\psi_s$ and $P_\hbar\psi_s$. Equations (2.4) give the diagonal entries. The cross moment (2.5) is purely imaginary, as is its complex conjugate, so both off-diagonal entries vanish. Diagonal multiplication gives $\widetilde{\gamma} = I_2$. Proposition 1.6, with $a = b = 1$, proves admissibility. □

The matrix T converts position and momentum to the dimensionless quantities X/s and $sP_\hbar/\hbar$. The covariance before this change of units retains $\hbar$; we express the admissibility criterion of Section 1 in dimensionless coordinates. With $\hbar = 1$ and $t = s^2$, the original covariance is $D(t, t^{-1})$, realizing the diagonal equality cases above.

Corollary 2.7. Lean ✓ *Let $m, \omega, \hbar > 0$ and set $s = \sqrt{\hbar/(m\omega)}$. Then*

$$\gamma_{\psi_s} = \operatorname{diag}\left(\frac{\hbar}{m\omega}, \hbar m\omega\right).$$

Proof. Positivity gives $s > 0$ and $s^2 = \hbar/(m\omega)$. Substitution into Theorem 2.6 gives the formula. □

3 Contraction of covariance matrices

We now turn to constructing Gaussian matrix product states from local Gaussian resources [3, Secs. 3 and 6]. At each site, we distinguish *physical modes*, which form the final system, from auxiliary *virtual modes*. Neighbouring sites are joined by EPR measurements on their virtual modes, and we seek the covariance of the physical modes conditioned on the measurement outcomes. At the covariance level, this construction leads to a Schur complement. We first establish the general matrix result, then explain EPR measurements and apply it to a periodic chain.

Theorem 3.1. Lean ✓ *Every admissible $2n\times 2n$ covariance matrix is positive definite: $x^T\gamma x > 0$ for every nonzero $x \in \mathbb{R}^{2n}$.*

Proof. Put $H = \gamma_{\mathbb{C}} - i\Omega_n$, where the subscript denotes entrywise inclusion in $\mathbb{C}$. For real x, skew-symmetry gives $x^T\Omega_n x = 0$, and hence

$$x^* H x = x^T \gamma x \geq 0.$$

For a positive semidefinite Hermitian matrix, a vector has zero quadratic form precisely when it lies in the kernel. Thus, if $x^T\gamma x = 0$, then $Hx = 0$. Taking imaginary parts gives $\Omega_n x = 0$. Since $\Omega_n^2 = -I$, we obtain $x = 0$. The quadratic form is therefore strictly positive away from zero. When $n = 0$, there is no nonzero vector and the assertion is vacuous. □

Definition 3.2 (Contracted covariance)**.** Let W be a real $(k + 2m) \times 2n$ matrix. For a real symmetric $2n \times 2n$ matrix γ, write

$$V = W\gamma W^T = \begin{pmatrix} E & F \\ F^T & G \end{pmatrix}, \qquad E \in \mathbb{R}^{k\times k}.$$

The first k output coordinates are to be eliminated; the last $2m$ are retained in position-then-momentum order. When E is invertible, define

$$\gamma_{\text{out}} = G - F^T E^{-1} F.$$

This is the *Schur complement* of E in V. For $k = 0$, it reduces to G.

The eliminated coordinates need not form a canonical subsystem. The next condition makes their symplectic pairings vanish, both among themselves and with the retained coordinates.

Theorem 3.3. Lean ✓ *Let $n, m, k \geq 0$ and let γ be admissible. Suppose W^T is one-to-one and*

$$W\Omega_n W^T = \begin{pmatrix} 0_{k\times k} & 0 \\ 0 & \Omega_m \end{pmatrix}.$$

Then E is positive definite, hence invertible, and γ_{out} is admissible.

Proof. Theorem 3.1 and injectivity of W^T give, for $y \neq 0$,

$$y^T V y = (W^T y)^T \gamma (W^T y) > 0.$$

Thus V and its principal block E are positive definite. Congruence of the input uncertainty matrix yields

$$Q = W_{\mathbb{C}}(\gamma_{\mathbb{C}} - i\Omega_n)W_{\mathbb{C}}^* = \begin{pmatrix} E_{\mathbb{C}} & F_{\mathbb{C}} \\ F_{\mathbb{C}}^* & G_{\mathbb{C}} - i\Omega_m \end{pmatrix} \succeq 0.$$

For any complex vector z, choose the eliminated component to cancel the first block row. Direct block multiplication gives

$$\begin{pmatrix} -E_{\mathbb{C}}^{-1}F_{\mathbb{C}}z \\ z \end{pmatrix}^* Q \begin{pmatrix} -E_{\mathbb{C}}^{-1}F_{\mathbb{C}}z \\ z \end{pmatrix} = z^*\big((G - F^T E^{-1} F)_{\mathbb{C}} - i\Omega_m\big)z \geq 0.$$

Here complexification commutes with the inverse of the invertible real matrix E. The output is real symmetric because E and G are symmetric and $(E^{-1})^T = E^{-1}$. Its uncertainty matrix is consequently Hermitian, so the quadratic-form inequality proves admissibility. The argument includes empty eliminated or retained coordinate sets. □

3.1 EPR measurements and covariance contraction

Two pairs of canonical quadratures admit the commuting combinations

$$X_- = q_A - q_B, \qquad P_+ = p_A + p_B, \qquad [X_-, P_+] = [q_A, p_A] - [q_B, p_B] = 0.$$

An *EPR measurement* is the joint measurement of these two observables. Their exact joint eigenvectors are generalized eigenvectors: they do not belong to the two-mode Hilbert space. In particular, the ideal EPR vector is not a normalizable state. Finite squeezing gives normalizable approximations, but we must not treat the infinitely squeezed vector as a density operator of trace one. Giedke and Cirac [2, Secs. III–IV] develop the covariance description of Gaussian measurements.

Reversing both quadratures of B is a canonical phase rotation. It replaces the preceding pair by

$$Y_q = q_A + q_B, \qquad Y_p = p_A - p_B. \tag{3.1}$$

We use this convention for the virtual pairs below. Put $\Theta = \mathrm{diag}(I_M, -I_M)$ for M such pairs. If we denote the remaining modes by C, the matrix

$$L = \begin{pmatrix} I & \Theta & 0 \\ 0 & 0 & I_C \end{pmatrix}$$

maps the quadrature vector in A, B, C order to (Y, C). The identity $\Theta\Omega_M\Theta^T = -\Omega_M$ gives $L\Omega L^T = \mathrm{diag}(0, \Omega_C)$, as required in Theorem 3.3.

We now compute the corresponding characteristic function, including its normalization. For a centered Gaussian with covariance γ, write

$$\chi(t) = \langle e^{it^T R} \rangle = \exp(-t^T \gamma t/4).$$

The factor $1/4$ follows from our convention that γ is twice the ordinary covariance. Thus the joint characteristic function of Y and the retained quadratures is $\chi(L^T(u, z))$. In A, B, C coordinates, $L^T(u, z) = (u, \Theta u, z)$: this explains the momentum sign in the contraction of characteristic-function variables. Write

$$L\gamma L^T = \begin{pmatrix} E & F \\ F^T & G \end{pmatrix}.$$

Conditioning on the zero outcome integrates out the Fourier variable u. The quadratic expression in the exponent is

$$\begin{aligned} u^T E u + 2u^T F z + z^T G z = (u + E^{-1}Fz)^T E (u + E^{-1}Fz) \\ + z^T (G - F^T E^{-1} F) z. \end{aligned}$$

Since E is positive definite, the Gaussian integral is finite. Dividing by its value at $z = 0$ leaves the characteristic function with covariance $G - F^T E^{-1} F$. More generally, for means d_Y, d_C and outcome a, the usual Gaussian conditioning formula is

$$d_{C|a} = d_C + F^T E^{-1}(a - d_Y), \qquad \gamma_{C|a} = G - F^T E^{-1} F. \tag{3.2}$$

The outcome changes the mean, but not the conditional covariance. We understand conditioning here through the outcome density, not as an event of positive probability at one exact value. These formulas give the measurement interpretation; Theorem 3.3 establishes the covariance inequality directly.

In the Choi–Jamiołkowski construction, the resource associated with a map is obtained formally by applying the map to one half of an ideal EPR pair. EPR contraction with the other half represents feeding an input to the map. This turns the construction of Gaussian matrix product states into contractions of local resources, but still involves ideal EPR measurements.

Moreover, a channel representative obtained from an infinitely squeezed pair need not be a finite covariance matrix: the identity channel already illustrates the difficulty. The results below use finite admissible local matrices. Schuch, Wolf and Cirac describe the resource construction in Sections 3 and 6 of [3]; their Section 5 replaces infinitely entangled bonds by finite ones while moving the infinite entanglement into the local maps.

3.2 A periodic chain

At each site $i \in \mathbb{Z}/N\mathbb{Z}$, we introduce M left virtual modes A_i, M right virtual modes B_i, and one retained mode C_i. The pair on each edge joins B_{i-1} to A_i. We assume $N \geq 3$ and $M \geq 1$. The integer M counts the *virtual modes per bond*. Each bosonic mode has an infinite-dimensional Hilbert space, so even $M = 1$ can support entanglement. With no virtual modes, $M = 0$, the virtual Hilbert space is one-dimensional; this is the counterpart of bond dimension $D = 1$ for a qudit MPS. We can also represent product covariances with $M \geq 1$ by adding unused virtual modes. A bound on M alone does not bound entanglement entropy: the squeezing must also be controlled.

Definition 3.4 (Periodic covariance)**.** Put $d = 2M + 1$ and let Γ_i be real $2d \times 2d$ matrices. The local mode order is $(A_{i,1}, \ldots, A_{i,M}, B_{i,1}, \ldots, B_{i,M}, C_i)$, with positions preceding momenta. Define γ_{loc} on Nd modes by

$$(\gamma_{\mathrm{loc}})_{(u,i,a),(v,j,b)} = \begin{cases} (\Gamma_i)_{(u,a),(v,b)}, & i = j, \\ 0, & i \neq j, \end{cases} \qquad u, v \in \{q, p\}.$$

Globally, positions precede momenta; within either group, sites precede local modes. The real linear map $W : \mathbb{R}^{2Nd} \to \mathbb{R}^{2NM} \oplus \mathbb{R}^{2N}$ has eliminated coordinates

$$(Wx)_{q,i,a} = x_{q,A_i,a} + x_{q,B_{i-1},a}, \qquad (Wx)_{p,i,a} = x_{p,A_i,a} - x_{p,B_{i-1},a},$$

and retained coordinates $(Wx)_{q,C_i} = x_{q,C_i}$ and $(Wx)_{p,C_i} = x_{p,C_i}$. Each output group is in position-then-momentum order. When its eliminated block is invertible, the *periodic covariance* is

$$\gamma_{\mathrm{per}} = \mathrm{SC}_{\mathbb{R}^{2NM}}(W \gamma_{\mathrm{loc}} W^T),$$

where SC denotes the Schur complement of Definition 3.2.

For comparison with [3], we group all A coordinates together, then all B and all C coordinates. If R shifts the site index to the right, $(Rx)_i = x_{i-1}$, the same map is

$$W = \begin{pmatrix} I_A & R\Theta_B & 0 \\ 0 & 0 & I_C \end{pmatrix}.$$

The shift acts identically on positions and momenta and preserves each mode label. This is the matrix in Section 6, Eq. (3) of [3].

Theorem 3.5. Lean ✓ *Suppose every Γ_i is admissible. Then W^T is injective,*

$$W \Omega_{Nd} W^T = \mathrm{diag}(0_{2NM}, \Omega_N),$$

the eliminated block of $W \gamma_{\mathrm{loc}} W^T$ is positive definite, and γ_{per} is admissible.

Proof. The independent sum of the local covariances is admissible by Theorem 1.5, iterated over the sites. Reordering its positions and momenta together gives γ_{loc} and preserves the uncertainty inequality. For an output vector (y_q, y_p, z_q, z_p), the A components of its image under W^T are y_q, y_p, and the C components are z_q, z_p. Hence W^T is injective. Each eliminated position–momentum pairing has an A contribution $+1$ and a B contribution -1. They cancel, including after the common cyclic shift. Distinct mode pairs and all pairings with C vanish. The retained pairings form Ω_N. Apply Theorem 3.3. □

3.3 Three correlated sites

Proposition 3.6. Lean ✓ *Take $N = 3$, $M = 1$, and at each site let*

$$K = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} I_3 & 0 \\ K & I_3 \end{pmatrix}, \qquad \Gamma_i = SS^T.$$

Then S is symplectic, Γ_i is admissible, and the periodic covariance is

$$\gamma_{\mathrm{per}} = \mathrm{diag}(Q, P), \qquad Q = \begin{pmatrix} 3/5 & 1/5 & 1/5 \\ 1/5 & 3/5 & 1/5 \\ 1/5 & 1/5 & 3/5 \end{pmatrix}, \quad P = \begin{pmatrix} 2 & -1/2 & -1/2 \\ -1/2 & 2 & -1/2 \\ -1/2 & -1/2 & 2 \end{pmatrix}.$$

It is admissible. In particular, distinct sites have position–position covariance entry $1/5$ and momentum–momentum covariance entry $-1/2$.

Proof. Since $K = K^T$, direct multiplication gives $S\Omega_3 S^T = \Omega_3$. The covariance I_6 is an independent sum of three one-mode vacua, so Theorem 1.2 gives admissibility of SS^T. Let R be the three-site cyclic right shift and U the 3×3 all-ones matrix. Multiplying the local covariance blocks gives

$$E = \begin{pmatrix} 2I & 0 \\ 0 & 4I - R - R^T \end{pmatrix}, \quad F = \begin{pmatrix} 0 & I + R \\ I - R & 0 \end{pmatrix}, \quad G = \begin{pmatrix} I & 0 \\ 0 & 3I \end{pmatrix}.$$

Here $R + R^T = U - I$ and $U^2 = 3U$. Consequently

$$(4I - R - R^T)^{-1} = (5I - U)^{-1} = \tfrac{1}{5}I + \tfrac{1}{10}U,$$

as multiplication checks. Since $U(I - R) = 0$, the two diagonal blocks of the Schur complement are

$$Q = I - (I - R)^T(\tfrac{1}{5}I + \tfrac{1}{10}U)(I - R) = \tfrac{2}{5}I + \tfrac{1}{5}U,$$
$$P = 3I - \tfrac{1}{2}(I + R)^T(I + R) = \tfrac{5}{2}I - \tfrac{1}{2}U.$$

The off-diagonal blocks vanish. Theorem 3.5 proves admissibility. □

For this example, $QP = I_3$, so the associated Gaussian state is pure. Its single-site covariance is $\mathrm{diag}(3/5, 2)$, with determinant $6/5 > 1$; the single-site reduced state is therefore mixed. Thus the state is entangled across every partition of one site from the other two. These deductions use the Gaussian purity criterion [1, Secs. II.C–D], which is cited, not proved here; Proposition 3.6 establishes the displayed covariance and its admissibility.

4 Homogeneous periodic covariances

Suppose that every site in the periodic construction uses the same local covariance Γ. A cyclic shift then leaves the construction unchanged. The finite Fourier transform separates its spatial frequencies: at each frequency, we obtain the covariance by eliminating only the local virtual coordinates. This gives the finite-chain form of the Fourier calculation in [3, Sec. 6].

Fix $N \geq 3$ and $M \geq 1$, with one physical mode at each site. We retain the ordering and predecessor wiring of Section 3. For the local block calculations, we group the coordinates as (a, b, c), where $a = (q_A, p_A)$, $b = (q_B, p_B)$ and $c = (q_C, p_C)$. Write

$$\Gamma = \begin{pmatrix} \Gamma_{vv} & \Gamma_{vC} \\ \Gamma_{Cv} & \Gamma_{CC} \end{pmatrix}, \qquad v = (a, b), \qquad \Theta = \mathrm{diag}(I_M, -I_M).$$

Thus the virtual block has size $4M \times 4M$, the cross block has size $4M \times 2$, and the physical block has size 2×2. All blocks are taken in the order (a, b, c).

Definition 4.1 (Fourier blocks)**.** Label sites and frequencies by $0, \dots, N-1$, with indices interpreted modulo N. Set

$$\phi_k = \frac{2\pi k}{N}, \qquad \widehat{f}_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{-\mathrm{i}\phi_k j} f_j, \qquad Z_k = \begin{pmatrix} I_{2M} & e^{-\mathrm{i}\phi_k}\Theta \end{pmatrix}.$$

Define

$$E_k = Z_k \Gamma_{vv} Z_k^*, \qquad F_k = Z_k \Gamma_{vC}, \qquad S_k = \Gamma_{CC} - F_k^* E_k^{-1} F_k, \tag{4.1}$$

where $*$ denotes conjugate transpose. Invertibility of E_k follows from the next theorem.

The eliminated coordinates are $y_j = a_j + \Theta b_{j-1}$. Substituting $j = r+1$ in the Fourier sum gives

$$\widehat{(b_{j-1})}_k = e^{-\mathrm{i}\phi_k} \widehat{b}_k, \qquad \widehat{y}_k = Z_k \begin{pmatrix} \widehat{a}_k \\ \widehat{b}_k \end{pmatrix}.$$

Theorem 4.2 (Fourier contraction)**.** Lean ✓ *Let Γ be admissible and let γ_{per} be its homogeneous periodic covariance. Each E_k is Hermitian positive definite, and*

$$(\gamma_{\mathrm{per}})_{\alpha,i+1;\,\beta,j+1} = (\gamma_{\mathrm{per}})_{\alpha i,\beta j} \qquad (\alpha, \beta \in \{q, p\}).$$

Its Fourier transform is block diagonal:

$$\frac{1}{N} \sum_{i,j=0}^{N-1} e^{-\mathrm{i}\phi_k i} (\gamma_{\mathrm{per}})_{\alpha i,\beta j} e^{\mathrm{i}\phi_l j} = \begin{cases} (S_k)_{\alpha\beta}, & k = l, \\ 0, & k \neq l. \end{cases} \tag{4.2}$$

Proof. For r coordinates at each site, define $T_k^{(r)} : \mathbb{C}^r \to \mathbb{C}^{Nr}$ by $(T_k^{(r)} v)_j = e^{\mathrm{i}\phi_k j} v$. Character orthogonality gives

$$(T_l^{(r)})^* T_k^{(r)} = N \delta_{lk} I_r, \qquad \frac{1}{N} \sum_k T_k^{(r)} (T_k^{(r)})^* = I_{Nr}.$$

The independent local covariance acts as Γ on each such subspace. The shift calculation above shows that the wiring acts there as $L_k = \mathrm{diag}(Z_k, I_2)$. Its transpose acts as L_k^*, since the contribution from a B coordinate uses the successor site. Consequently the wired covariance has local matrix

$$L_k \Gamma L_k^* = \begin{pmatrix} E_k & F_k \\ F_k^* & \Gamma_{CC} \end{pmatrix}.$$

Write its global blocks as E, F, G, as in the contraction theorem. In particular,

$$E T_k^{(2M)} = T_k^{(2M)} E_k, \qquad F T_k^{(2)} = T_k^{(2M)} F_k, \qquad G T_k^{(2)} = T_k^{(2)} \Gamma_{CC}.$$

Theorem 3.1 gives $\Gamma \succ 0$ over the reals. For complex vectors, put $Q = \Gamma - \mathrm{i}\Omega$, with the symplectic matrix Ω expressed in the same coordinate order. Both Q and Q^T are positive semidefinite, so their average Γ is positive semidefinite as well. Since Γ is nonsingular, it is positive definite. Its virtual principal block is therefore positive definite. The map Z_k^* is injective, since its first block is the identity, and hence $E_k = Z_k \Gamma_{vv} Z_k^* \succ 0$. By Theorem 3.5, the global block E is also positive definite. The first mode identity therefore implies $E^{-1} T_k^{(2M)} = T_k^{(2M)} E_k^{-1}$. Taking the Schur complement yields

$$\gamma_{\mathrm{per}} T_k^{(2)} = T_k^{(2)} S_k.$$

Multiplication by $(T_l^{(2)})^*$ and orthogonality prove (4.2). A cyclic shift acts by a scalar on each Fourier subspace, so γ_{per} commutes with cyclic shifts and is translation invariant. □

Theorem 4.3 (Reconstruction and uncertainty)**.** Lean ✓ *Under the same assumptions,*

$$(\gamma_{\mathrm{per}})_{\alpha i,\beta j} = \frac{1}{N}\sum_{k=0}^{N-1} e^{\mathrm{i}\phi_k(i-j)}(S_k)_{\alpha\beta}. \tag{4.3}$$

Moreover,

$$S_{-k} = \overline{S_k}, \qquad S_k - \mathrm{i}\Omega_1 \succeq 0, \qquad \Omega_1 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

where $-k$ is interpreted modulo N and the bar is entrywise conjugation.

Proof. Insert the Fourier resolution of the identity into $\gamma_{\mathrm{per}} T_k^{(2)} = T_k^{(2)} S_k$ to obtain (4.3). Since γ_{per} is real, conjugating its Fourier transform replaces each frequency by its negative. The diagonal blocks in (4.2) therefore satisfy $S_{-k} = \overline{S_k}$. Finally, Theorem 3.5 gives $\gamma_{\mathrm{per}} - \mathrm{i}\Omega_N \succeq 0$. Its pullback along $T_k^{(2)}$ is $N(S_k - \mathrm{i}\Omega_1)$: the position and momentum coordinates use the same Fourier vectors, whose squared norm is N. Dividing by $N > 0$ proves the inequality. □

The matrices S_k can have complex entries. Conjugate frequencies together reconstruct the real covariance; the uncertainty inequality for an individual frequency is an inequality of complex Hermitian matrices.

5 Product covariances

The periodic construction includes products: if the physical mode at each site is independent of the virtual modes, contraction leaves its covariance unchanged. We make this explicit, allowing the physical covariance to vary from site to site. We retain $N \geq 3$, $M \geq 1$ and the ordering of Section 3.

Definition 5.1. Let η_i be a real one-mode covariance at site i. Take the local covariance

$$\Gamma_i = \mathrm{diag}(I_{2M}, I_{2M}, \eta_i)$$

in grouped phase coordinates $(q_{A_i}, p_{A_i}, q_{B_i}, p_{B_i}, q_{C_i}, p_{C_i})$, then reorder it to the position-then-momentum coordinates used for periodic assembly. Define the physical product covariance by

$$(\gamma_{\mathrm{prod}})_{(u,i),(v,j)} = \begin{cases} (\eta_i)_{uv}, & i = j, \\ 0, & i \neq j, \end{cases} \qquad u, v \in \{q, p\}.$$

Each virtual mode is a vacuum and is independent of all other modes at its site. The physical mode has covariance η_i.

Theorem 5.2. Lean ✓ *If every η_i is admissible, then every Γ_i is admissible and*

$$W\gamma_{\mathrm{loc}}W^T = \begin{pmatrix} 2I_{2NM} & 0 \\ 0 & \gamma_{\mathrm{prod}} \end{pmatrix}.$$

Consequently the periodic contraction equals γ_{prod} and is admissible. In particular, one virtual mode per bond suffices for every $N \geq 3$.

Proof. The independent-sum rule of Theorem 1.5 applies to the local resources. In grouped phase coordinates its uncertainty matrix is

$$\Gamma_i - i\,\mathrm{diag}(\Omega_M, \Omega_M, \Omega_1) = (I_{2M} - i\Omega_M) \oplus (I_{2M} - i\Omega_M) \oplus (\eta_i - i\Omega_1) \succeq 0.$$

The two vacuum blocks and the prescribed physical block are positive semidefinite; reordering the coordinates preserves the inequality.

Each eliminated quadrature is the signed sum of two independent virtual vacuum coordinates. Its diagonal covariance entry is $1 + 1 = 2$. Distinct eliminated coordinates have zero covariance: distinct virtual modes are independent, and within each vacuum mode the position–momentum covariance vanishes. The virtual coordinates are also independent of every physical coordinate. Thus, in the notation of Definition 3.2,

$$E = 2I_{2NM}, \qquad F = 0, \qquad G = \gamma_{\text{prod}}.$$

The Schur-complement formula therefore gives

$$\gamma_{\text{per}} = G - F^T E^{-1} F = \gamma_{\text{prod}}.$$

Admissibility follows from Theorem 3.5. □

Corollary 5.3. Lean ✓ *For $\eta_i = I_2$ at every site, the periodic contraction is I_{2N}. For arbitrary $s_i > 0$, the choices $\eta_i = \operatorname{diag}(s_i^2, s_i^{-2})$ give the corresponding product of squeezed one-mode covariances.*

Proof. The vacuum covariance is admissible. The diagonal criterion of Proposition 1.6 states

$$\operatorname{diag}(a, b) \text{ is admissible} \iff a > 0 \text{ and } ab \geq 1.$$

Apply it with $a = s_i^2$ and $b = s_i^{-2}$, whose product is one, and use Theorem 5.2. In the vacuum case, the entries of γ_{prod} are those of the identity. □

The one-mode Gaussians of Theorem 2.6, with $\hbar = 1$, have these squeezed covariances. The virtual vacua in this construction are unused; adding them allows products to fit the same $M \geq 1$ convention as the correlated examples.

6 The formal Fock space and the Bargmann picture

The number basis of a bosonic Fock space labels vectors by occupation numbers. Allowing all coefficient families, with no summability condition, gives a space on which the creation operators and the exponentials of operators that raise the particle number act exactly, one coefficient at a time. This section develops that formal Fock space, the canonical commutation relations on it, the exponential law for raising operators, and the Bargmann picture, in which creation and annihilation become multiplication by and differentiation in formal variables. The exponentials of the Gaussian-augmented bosonic matrix product states of Tjoa and Cirac [4] are of this kind. Those states are a different construction from the Gaussian matrix product states whose covariance matrices are built by contraction in Sections 3–5; these notes do not yet relate the two.

Notation for this section.

ι	a set of modes
$n = (n_i)_{i\in\iota}$	a multi-index of occupation numbers, finitely many nonzero
m	an occupation number of a single mode
e_i	the multi-index with a single 1 at i
$\lvert n\rvert = \sum_i n_i$	total particle number of n
$n! = \prod_i n_i!$	factorial of a multi-index
$\lvert n\rangle$	number state (Definition 6.1)
$\mathcal{F}(\iota, R)$	formal Fock space with coefficients in R (Definition 6.1)
$\mathfrak{h}$, $\Gamma_s(\mathfrak{h})$	for finite ι, the one-particle space $\mathbb{C}^\iota$ and its Fock space
$a_i^\dagger$, a_i	creation and annihilation operators (Definition 6.2)
T, T', e^T	raising operators and their exponential (Definition 6.4)
$R[[z]]$, ∂_j	formal power series in $(z_i)_{i\in\iota}$ over R; partial derivative
$[z^n]\, f$	coefficient of $z^n = \prod_i z_i^{n_i}$ in $f \in R[[z]]$
$\mathfrak{B}$	Bargmann map (Definition 6.7)

In this section R is a $\mathbb{C}$-algebra, $X \in R$, n a multi-index, W a word in the ladder operators, T a raising operator, $p \geq 1$ an exponent, z the formal variables and Λ a bound on particle numbers; letters shared with Sections 1–5 do not carry their meanings from there.

6.1 The formal Fock space

Throughout, R is a $\mathbb{C}$-algebra: a ring together with a ring homomorphism from $\mathbb{C}$ into its centre. The two cases used later are $R = \mathbb{C}$ (states) and $R = M_\beta(\mathbb{C})$, the complex matrices indexed by a finite set β (vectors that carry a matrix index); for β empty, $M_\beta(\mathbb{C})$ is the zero ring, which is allowed.

Definition 6.1 (Formal Fock space)**.** $\mathcal{F}(\iota, R)$ is the set of all functions ψ from multi-indices to R, with pointwise addition and scalar multiplication; ψ is read as $\sum_n \psi(n)|n\rangle$. The number state $c|n\rangle$ has coefficient c at n and 0 elsewhere, and $|0\rangle$ is the vacuum. A vector ψ *vanishes below* k if $\psi(n) = 0$ whenever $|n| < k$.

No summability is imposed. For finite ι and the one-particle space $\mathfrak{h} = \mathbb{C}^\iota$ with its standard orthonormal basis, the k-particle space $\otimes_s^k \mathfrak{h}$ has as orthonormal basis the number states $|n\rangle$ with $|n| = k$, so that, sector by sector,

$$\Gamma_{\text{fin}}(\mathfrak{h}) = \bigoplus_k \otimes_s^k \mathfrak{h} \;\subset\; \Gamma_s(\mathfrak{h}) = \overline{\bigoplus_k} \otimes_s^k \mathfrak{h} \;\subset\; \prod_k \otimes_s^k \mathfrak{h} = \mathcal{F}(\iota, \mathbb{C}) :$$

the finite-particle space, the bosonic Fock space of [5, 6] (the Hilbert direct sum, that is, the square-summable families) and the full product. $\Gamma_s(\mathfrak{h})$ is recalled from [5, 6]; it is not constructed in these notes, nor are second quantization, the Fock representation of the CCR algebra, or the

identification of $\Gamma_s(\mathfrak{h})$ with $L^2(\mathbb{R}^\iota)$ through the Hermite functions that would join this section to Sections 1–5. For a single mode, the vector with coefficients $\sqrt{(2k)!}/k!$ at $|2k\rangle$ and 0 at odd occupations lies in $\mathcal{F}(\{1\},\mathbb{C})$ but not in Γ_s, since $(2k)!/(k!)^2 \geq 1$. After Definition 6.4 it is $e^{(a^\dagger)^2}|0\rangle$.

Definition 6.2 (Ladder operators and left multiplication)**.** For $i \in \iota$, the creation and annihilation operators are the $\mathbb{C}$-linear maps

$$(a_i^\dagger \psi)(n) = \begin{cases} \sqrt{n_i}\,\psi(n-e_i), & n_i \geq 1, \\ 0, & n_i = 0, \end{cases} \qquad (a_i\psi)(n) = \sqrt{n_i+1}\,\psi(n+e_i).$$

For $X \in R$, $X\psi$ is the vector with coefficients $X\psi(n)$. For a word W in the ladder operators, $X \otimes W$ denotes the composite XW; the two factors commute, since a ladder word multiplies each coefficient by complex scalars and moves it, while X multiplies it on the left, and complex scalars are central in R.

On number states these are the formulas $a_i^\dagger|n\rangle = \sqrt{n_i+1}|n+e_i\rangle$ and $a_i|n\rangle = \sqrt{n_i}|n-e_i\rangle$. On $\Gamma_s(\mathfrak{h})$ the creation and annihilation operators of [5] are $a^*(h) = \sum_i h_i a_i^\dagger$ and $a(h) = \sum_i \overline{h_i}\, a_i$ for $h \in \mathfrak{h}$. Each coefficient of $a_i^\dagger\psi$ or $a_i\psi$ involves a single coefficient of ψ, so both operators are defined on the whole formal Fock space.

Theorem 6.3 (Canonical commutation relations)**.** Lean ✓ *On all of $\mathcal{F}(\iota, R)$: $[a_i, a_j^\dagger] = \delta_{ij}$, $[a_i, a_j] = 0$ and $[a_i^\dagger, a_j^\dagger] = 0$.*

Proof. We check the identities coefficientwise.

(a) For $i = j$: $(a_i a_i^\dagger\psi)(n) = \sqrt{n_i+1}\sqrt{n_i+1}\,\psi(n)$ and $(a_i^\dagger a_i\psi)(n) = \sqrt{n_i}\sqrt{n_i}\,\psi(n)$, the second being 0 when $n_i = 0$. The difference is $\psi(n)$.

(b) For $i \neq j$, both orders give $\sqrt{n_j}\sqrt{n_i+1}\,\psi(n+e_i-e_j)$ when $n_j \geq 1$, and 0 otherwise, because shifting n_i does not change n_j.

(c) For $i \neq j$, $a_i a_j$ and $a_j a_i$ both multiply $\psi(n+e_i+e_j)$ by $\sqrt{n_i+1}\sqrt{n_j+1}$; $a_i^\dagger a_j^\dagger$ and $a_j^\dagger a_i^\dagger$ both multiply $\psi(n-e_i-e_j)$ by $\sqrt{n_i}\sqrt{n_j}$ when $n_i, n_j \geq 1$, and give 0 otherwise. For $i = j$ there is nothing to prove. □

Since $a_i^\dagger(c|n\rangle) = \sqrt{n_i+1}\,c|n+e_i\rangle$ and the creation operators commute, $\prod_i (a_i^\dagger)^{n_i}(c|0\rangle) = \sqrt{n!}\,c|n\rangle$, the product running over the finitely many i with $n_i \neq 0$. So $|n\rangle = \prod_i (a_i^\dagger)^{n_i}|0\rangle/\sqrt{n!}$, as usual.

6.2 Raising operators and their exponentials

Operators that raise the particle number have exponential series that are finite at each coefficient.

Definition 6.4 (Raising operator and its exponential)**.** A $\mathbb{C}$-linear map T on $\mathcal{F}(\iota, R)$ is *raising* (short for order-raising) if, for every k, $T\psi$ vanishes below $k+1$ whenever ψ vanishes below k. For raising T,

$$(e^T\psi)(n) = \sum_{r=0}^{|n|} \frac{(T^r\psi)(n)}{r!}.$$

Lemma 6.5. Lean ✓ *Let T and T' be raising.*

(a) If $T_1, \ldots, T_r$ are raising and ψ vanishes below k, then $T_1 \cdots T_r\psi$ vanishes below $k+r$. In particular $(T^r\psi)(n) = 0$ for $r > |n|$, and $(e^T\psi)(n) = \sum_{r=0}^{\Lambda}(T^r\psi)(n)/r!$ for every $\Lambda \geq |n|$.

(b) $(T\psi)(n)$ depends only on the coefficients $\psi(n')$ with $|n'| < |n|$.

(c) $T + T'$, TT' *and* cT $(c \in \mathbb{C})$ *are raising; so are* $a_i^\dagger$ *and, for* $X \in R$ *and* $p \geq 1$, $X(a_i^\dagger)^p$.

Proof. (a) Apply the definition to T_r, then T_{r-1}, and so on; each step raises the bound by one. Every ψ vanishes below 0.

(b) If ψ and ψ' agree in degrees below $|n|$, then $\psi - \psi'$ vanishes below $|n|$, so $T(\psi - \psi')$ vanishes below $|n| + 1$, in particular at n.

(c) The first three claims follow from the definition. $(a_i^\dagger \psi)(n)$ is a multiple of $\psi(n - e_i)$ and $|n - e_i| = |n| - 1$, so $a_i^\dagger$ is raising; hence so is $(a_i^\dagger)^p$ for $p \geq 1$, and left multiplication by X preserves vanishing below k. □

We define the exponential by the finite sum of Definition 6.4, not as a limit, so no topology on $\mathcal{F}(\iota, R)$ is needed. By Lemma 6.5(a) this is the operator series $\sum_r T^r/r!$ itself: at every coefficient the partial sums are constant from $r = |n|$ on. When the series converges in $\Gamma_s(\mathfrak{h})$, its limit has these coefficients, since convergence there implies convergence of each coefficient.

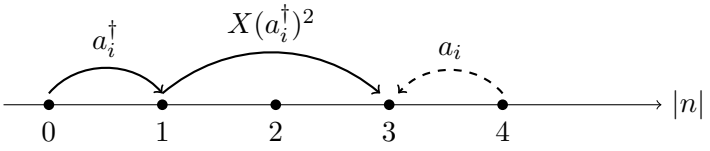


Figure 2: Raising maps (solid) move every coefficient to strictly higher particle number; the annihilation operator (dashed) moves it down. Hence the coefficient of $e^T\psi$ at particle number k receives only the terms $T^r\psi$ with $r \leq k$: the finite sum of Definition 6.4.

Theorem 6.6 (Exponential law). Lean ✓ *If* T *and* T' *are raising and* $TT' = T'T$, *then* $e^{T+T'} = e^T e^{T'}$.

Proof. By Lemma 6.5(c), $T + T'$ is raising, so both sides are defined; likewise T^r is raising for $r \geq 1$, which is used below. The coefficient of $e^T e^{T'}\psi$ at n reduces to a finite double sum, which the binomial theorem collapses. Fix n and ψ, and put $\Lambda = |n|$. By Lemma 6.5(a), $e^{T'}\psi$ agrees in all degrees $\leq \Lambda$ with $\sum_{t=0}^{\Lambda} T'^t\psi/t!$. By Lemma 6.5(b), applied to the raising map T^r for $r \geq 1$ (and trivially for $r = 0$),

$$(e^T e^{T'}\psi)(n) = \sum_{r=0}^{\Lambda}\sum_{t=0}^{\Lambda} \frac{(T^r T'^t\psi)(n)}{r!\,t!} = \sum_{k=0}^{\Lambda} \frac{1}{k!} \sum_{r+t=k} \binom{k}{r} (T^r T'^t\psi)(n) = \sum_{k=0}^{\Lambda} \frac{((T+T')^k\psi)(n)}{k!}.$$

We dropped the terms with $r + t > \Lambda$ because $T^r T'^t$ is a product of $r + t$ raising maps and therefore, by Lemma 6.5(a), vanishes at n; the last step is the binomial theorem for commuting T and T'. The right side is $(e^{T+T'}\psi)(n)$. □

For matrices the same identity rests on absolute convergence of the double series. Here the grading by particle number replaces it. Without the raising hypothesis the finite sum of Definition 6.4 need not agree with the series $\sum_r (T^r\psi)(n)/r!$, and the series need not converge. For the annihilation operator a of a single mode and $\psi = |1\rangle$, the finite sum at $n = 0$ is $\psi(0) = 0$, while the series gives $(a\psi)(0) = 1$. For $\psi(m) = m!$, the series $\sum_k (a^k\psi)(0)/k! = \sum_k \sqrt{k!}$ diverges.

6.3 The Bargmann picture

$R[[z]]$ is the set of all functions f from multi-indices to R, written $f = \sum_n [z^n]\, f\, z^n$, with the Cauchy product $[z^n]\,(fg) = \sum_{n'+n''=n} [z^{n'}]\, f\, [z^{n''}]\, g$, a finite sum. For $f \in R[[z]]$, we define the partial derivative coefficientwise by $[z^n]\,(\partial_j f) = (n_j + 1)\,[z^{n+e_j}]\, f$; for $R = \mathbb{C}$ it is the usual derivative of formal power series.

Definition 6.7 (Bargmann map)**.** $\mathfrak{B} : \mathcal{F}(\iota, R) \to R[[z]]$ is given by $[z^n]\,\mathfrak{B}\psi = \psi(n)/\sqrt{n!}$. It is a $\mathbb{C}$-linear bijection, with inverse $\psi(n) = \sqrt{n!}\,[z^n]\,f$.

Thus $\mathfrak{B}|n\rangle = z^n/\sqrt{n!}$, which for finite ι is the orthonormal basis of Bargmann's space of entire functions [7]. So $\mathfrak{B}$ maps $\Gamma_s(\mathfrak{h})$ onto the Taylor series of the Bargmann space, and the formal Fock space onto all of $R[[z]]$, including, for a single mode and $R = \mathbb{C}$, series such as $\sum_m m!\, z^m = \mathfrak{B}\psi$, $\psi(m) = m!\sqrt{m!}$, with radius of convergence 0. A vector ψ vanishes below k exactly when $\mathfrak{B}\psi$ has order at least k (no monomial of total degree below k), so a raising map raises the order of power series by at least one; hence the name.

Lemma 6.8. Lean ✓ *If $n_j \geq 1$, then $n! = n_j\,(n - e_j)!$.*

Proof. n and $n - e_j$ differ only in the j-th entry, and $n_j! = n_j\,(n_j - 1)!$. □

Lemma 6.9. Lean ✓ *For every $f \in R[[z]]$: $[z^n]\,(z_j^m f) = [z^{n-me_j}]\,f$ if $m \leq n_j$, and 0 if $m > n_j$.*

Proof. This is the coefficient formula for multiplication by the monomial z_j^m; it holds for non-commutative R because z_j^m has a scalar coefficient. □

Theorem 6.10 (Intertwining)**.** Lean ✓ *For every $\psi \in \mathcal{F}(\iota, R)$ and $X \in R$:*

(a) $\mathfrak{B}(a_j^\dagger \psi) = z_j\,\mathfrak{B}\psi$*;*

(b) $\mathfrak{B}(a_j \psi) = \partial_j \mathfrak{B}\psi$*;*

(c) $\mathfrak{B}(X\psi) = X\,\mathfrak{B}\psi$*;*

(d) $\mathfrak{B}|0\rangle = 1$.

Proof. (a) At n with $n_j \geq 1$, Lemma 6.8 gives

$$[z^n]\,\mathfrak{B}(a_j^\dagger \psi) = \frac{\sqrt{n_j}\,\psi(n - e_j)}{\sqrt{n!}} = \frac{\psi(n - e_j)}{\sqrt{(n - e_j)!}},$$

which is $[z^n]\,(z_j \mathfrak{B}\psi)$ by Lemma 6.9. If $n_j = 0$, both sides vanish.

(b) By Lemma 6.8 at $n + e_j$,

$$[z^n]\,\mathfrak{B}(a_j \psi) = \frac{\sqrt{n_j + 1}\,\psi(n + e_j)}{\sqrt{n!}} = (n_j + 1)\,\frac{\psi(n + e_j)}{\sqrt{(n + e_j)!}},$$

which is $[z^n]\,(\partial_j \mathfrak{B}\psi)$.

(c) $1/\sqrt{n!}$ is a complex scalar and commutes with X.

(d) $[z^0]\,\mathfrak{B}|0\rangle = 1/\sqrt{0!} = 1$, and all other coefficients vanish. □

For $X \in R$, $j \in \iota$ and $p \geq 1$, the formal exponential $e^{Xz_j^p}$ is the series $\sum_{k \geq 0} X^k z_j^{pk}/k!$ in the single variable z_j.

Theorem 6.11 (Exponentials in the Bargmann picture)**.** Lean ✓ *For $X \in R$, $j \in \iota$, $p \geq 1$ and every $\psi \in \mathcal{F}(\iota, R)$: $\mathfrak{B}\big(e^{X \otimes (a_j^\dagger)^p}\psi\big) = e^{Xz_j^p}\,\mathfrak{B}\psi$.*

Proof. Write $T = X(a_j^\dagger)^p$ and $f = \mathfrak{B}\psi$. By Theorem 6.10, and since z_j commutes with every series, induction on k gives $\mathfrak{B}(T^k\psi) = X^k z_j^{pk} f$. Fix n. Then

$$[z^n]\,\mathfrak{B}(e^T\psi) = \sum_{k=0}^{|n|} \frac{1}{k!}[z^n]\,(X^k z_j^{pk} f).$$

The coefficient of a product Pf at n involves only the coefficients of P at multi-indices $n' \le n$. For such n', the coefficient of $e^{Xz_j^p}$ at n' equals that of the polynomial $\sum_{k\le|n|} X^k z_j^{pk}/k!$: a term with $k > |n|$ sits at a multi-index of total degree $pk \ge k > |n| \ge |n'|$. Hence $[z^n]\,(e^{Xz_j^p} f)$ is the same finite sum. □

The truncation in Definition 6.4 therefore cuts only terms whose coefficient at n vanishes.

Index of terms

Each term is defined where it first appears; the number is that of the definition, or of the statement or section where the term is introduced. Terms marked * are introduced in these notes or used in a sense or normalization different from the usual one.

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Bargmann map Def. 6.7
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vacuum Def. 6.1
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virtual modes per bond* §3

References

[1] C. Weedbrook, S. Pirandola, R. García-Patrón, N. J. Cerf, T. C. Ralph, J. H. Shapiro and S. Lloyd, *Gaussian quantum information*, Reviews of Modern Physics **84**, 621 (2012). doi:10.1103/RevModPhys.84.621; arXiv:1110.3234.

[2] G. Giedke and J. I. Cirac, *The characterization of Gaussian operations and distillation of Gaussian states*, Physical Review A **66**, 032316 (2002). doi:10.1103/PhysRevA.66.032316; arXiv:quant-ph/0204085v4.

[3] N. Schuch, M. M. Wolf and J. I. Cirac, *Gaussian matrix product states*, in *Quantum Information and Many Body Quantum Systems*, edited by M. Ericsson and S. Montangero (Edizioni della Normale, Pisa, 2008), p. 129. arXiv:1201.3945v1 (2012).

[4] E. Tjoa and J. I. Cirac, *Gaussian-augmented bosonic matrix-product states: theory and applications*, arXiv:2607.28753 (2026).

[5] J. Dereziński and C. Gérard, *Mathematics of Quantization and Quantum Fields*, Cambridge Monographs on Mathematical Physics (Cambridge University Press, 2013).

[6] O. Bratteli and D. W. Robinson, *Operator Algebras and Quantum Statistical Mechanics 2*, 2nd ed. (Springer, Berlin, 1997).

[7] V. Bargmann, *On a Hilbert space of analytic functions and an associated integral transform. Part I*, Communications on Pure and Applied Mathematics **14**, 187 (1961). doi:10.1002/cpa.3160140303.

Lean declarations

The linked Lean ✓ tag marks a result whose Lean proof has been checked and whose correspondence with the statement here has been reviewed. Only marked statements are listed, each with the declarations of BosonicLean that prove it.

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